

Ayush Prakash Tiwari

Software Engineer | Full Stack Developer

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Education

Jaypee University of Engineering and Technology

B.Tech in Computer Science and Engineering

Aug 2023 – 2027

Guna, Madhya Pradesh

Summary

Software Engineer and Full Stack Developer with strong experience in **Java, Spring Boot, Python, SQL, and Microservices**. Skilled in designing scalable REST APIs, AI-powered applications, Retrieval-Augmented Generation (RAG) systems, and cloud-native solutions, with proven open-source contributions to the Eclipse Foundation and a passion for building high-performance, production-ready software.

Experience

Open Source Contributor | Eclipse Foundation

2026

- Designed and implemented a **Quick Export feature** for Thing Description (TD) affordances (actions, properties, events) in the **eclipse-editord** IoT platform, enabling **single-click export** and reducing manual steps in IoT workflow configuration – [PR #181](#).
- Improved **export process speed by 50%**, enhancing productivity for **1,000+ developers** and IoT professionals; delivered with **zero user-reported issues** post-release via Git workflows, pull requests, and code reviews.
- Collaborated with **maintainers and the open-source community** to review feedback, refine implementation details, and ensure compliance with project coding standards, documentation, and CI workflows.

Projects

Self-Healing RAG Pipeline | *Python, FastAPI, Java 21, Spring Boot, React, TypeScript, Groq AI, Qdrant, MongoDB, Docker* [Link](#)

- Architected a **3-stage self-healing retrieval pipeline** with escalating strategies ($k=4 \rightarrow 8 \rightarrow 12$), **automatic query rewriting**, and critic-based answer grounding validation to eliminate silent RAG failures on large PDF corpora.
- Built **Python FastAPI microservice** with `async/await` for concurrent users and **Java 21 Spring Boot** API gateway featuring **Google OAuth2, JWT security**, MongoDB user management, and conversational history tracking; engineered **React TypeScript** frontend with document upload and RAG interaction UI.
- Delivered **production-ready monorepo** with **Docker Compose** orchestration, **3-attempt retry logic** with progressive retrieval expansion, full JSON trace logging with confidence scores and source citations, and plug-and-play **LLM/vector store abstraction** ($\text{Groq} \leftrightarrow \text{Qdrant} \leftrightarrow \text{Pinecone} \leftrightarrow \text{ChromaDB}$).

Image Encryption Vault | *React 18.3, TypeScript 5.5, Vite, Tailwind CSS, shadcn/ui, Vitest*

[Link](#)

- Engineered a **client-side image encryption application** using **XOR-based pixel encryption** with deterministic key hashing – **zero server uploads**, no external requests, complete data privacy guaranteed by design.
- Implemented **batch processing** for simultaneous encrypt/decrypt of multiple images with **real-time progress tracking**; built custom components including drag-and-drop upload zone, secret key strength indicator, and **before/after comparison slider**.
- Deployed on **Vercel** with a **95.6% TypeScript codebase** ensuring full type safety; achieved **60fps smooth animations**, responsive design across desktop, tablet, and mobile with glassmorphism dark/light theme toggle.

Technical Skills

Programming Languages: Java, Python, C++, JavaScript, TypeScript, SQL

Technologies / Frameworks: Spring Boot, FastAPI, React.js, LangChain, TensorFlow, Keras, Scikit-learn, Librosa, OpenCV, NumPy, Pandas

Databases & Cloud: MongoDB, MySQL, Redis, Qdrant, Docker, REST APIs, Microservices, OAuth2, JWT

Developer Tools: Git, GitHub, Postman, Docker Compose

Concepts: Retrieval-Augmented Generation (RAG), Deep Learning, CNN, Microservices Architecture, Event-Driven Systems

Achievements

- Secretary**, Computer Society of India (CSI) – Led technical development and managed projects for university student chapter
- Runner-up** – Walmart Sparkathon, a national-level hackathon competing against **500+ teams** across India
- Full Stack Developer (Hackathon)** – Built 2 MERN platforms (500+ users) with JWT auth and REST APIs; achieved -35% load time and +40% DB speed; delivered with 99.9% uptime in 48 hrs (2024)

Certifications

- Introduction to Deep Learning – [Certificate Link](#)
- Artificial Intelligence Foundation Certification – [Certificate Link](#)